

Systematic Review of Academic Procrastination Interventions in Computing Higher Education

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Abstract

Academic procrastination is a persistent challenge in computing education, yet evidence on the effectiveness of course-level interventions remains fragmented across diverse designs and contexts. We present a systematic literature review of studies published in the past decade that empirically examine interventions to reduce academic procrastination among post-secondary computing students. Evidence from 19 articles examines interventions that target procrastination through structural, feedback-based, motivational, and self-regulatory mechanisms. Our findings suggest that interventions introducing clear temporal structure consistently promote earlier starts and more distributed work, which act as key mediators of performance gains. The magnitude of these gains depends strongly on task structure, with greater benefits for long-horizon, multi-step assignments than for short, routine tasks. Moreover, supportive designs reliably outperform punitive or restrictive schemes, while uniform interventions yield uneven benefits across students. This review highlights the importance of designing structured, supportive, and personalized interventions to address procrastination in computing education.

1 Introduction

Procrastination has been defined as the intentional delay of tasks despite knowing the potential negative consequences [1]. Academic procrastination is a pervasive issue affecting students across various disciplines, significantly hindering academic performance and personal well-being.

Students in computing programs face a distinct set of challenges that contribute to academic procrastination. For instance, many students postpone starting their work because a complete software development cycle, encompassing analysis, design, coding, testing, and documentation, can be unfamiliar and substantially more time-consuming than coursework in many other disciplines [2]. Beyond course structure, students often face significant mental health challenges, including high